Heart Disease Prediction Using Machine Learning

Comparative Analysis of Classification Algorithms

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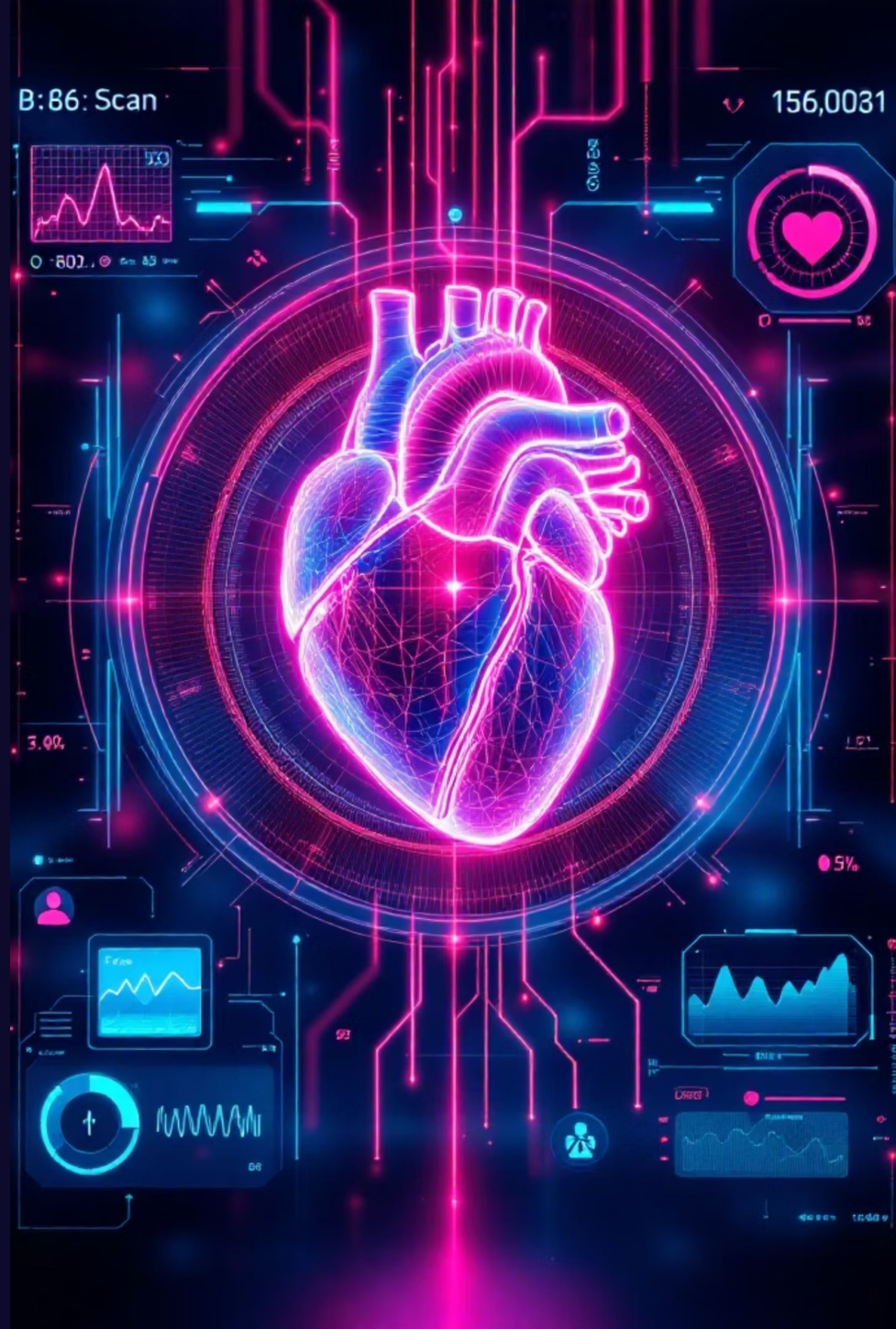


Project Overview & Research Question

Research Question

Can ML classification models accurately predict heart disease presence, and which algorithms and features perform best?

Dataset Challenge

Small size (n=303), class imbalance (54:46 ratio)

ML Challenge

Compare traditional vs ensemble methods for tabular data

Key Literature Benchmarks

- Detrano et al. (1989): Logistic Regression baseline at 77% accuracy
- Alizadehsani et al. (2013): Hybrid ML achieved 86.5% accuracy
- Probst et al. (2019): Random Forest tuning methodology

Research Gap & Unique Contribution

RESEARCH GAP:

- Limited reproducible feature importance studies on test sets
- No comprehensive hyperparameter tuning ROI comparison
- Lack of systematic ensemble vs traditional ML comparison

MY UNIQUE CONTRIBUTION:

- First test-set permutation importance validation on UCI dataset
- Quantifies hyperparameter tuning value (XGBoost +1.64%, RF 0%)
- Provides reproducible ML methodology for small medical datasets

Dataset & EDA Insights

UCI Cleveland Heart Disease Dataset (n=303 patients,x 13 features)

Class Distribution

Class 0 (Healthy): 164 samples (54.1%)

Class 1 (Diseased): 139 samples (45.9%)

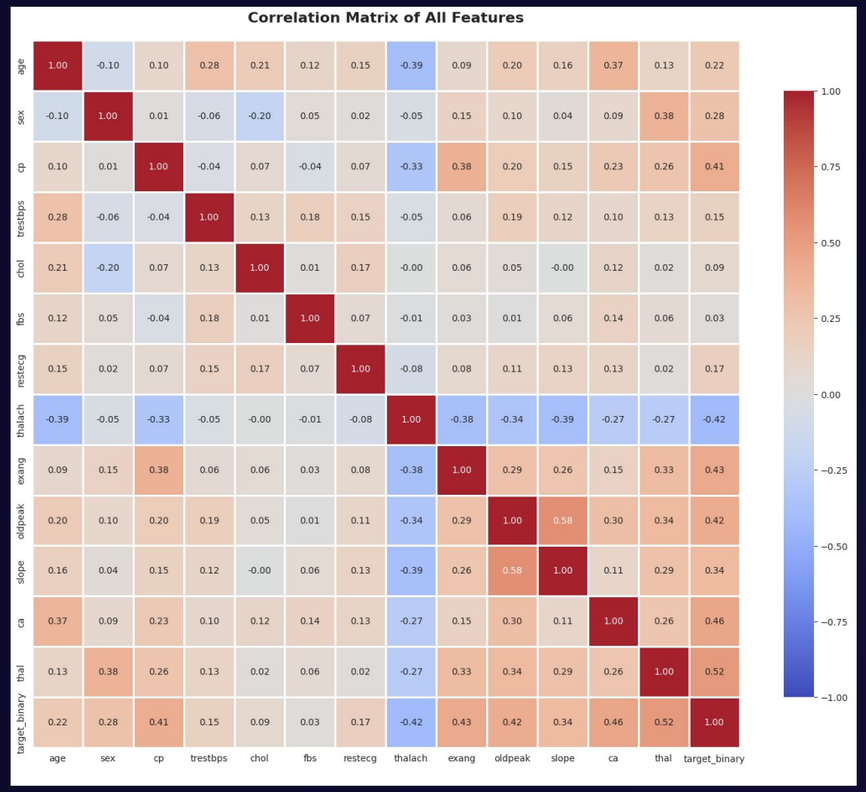
Moderate imbalance → stratified sampling used

Missing Values

ca: 4 missing (1.32%)

thal: 2 missing (0.66%)

Total: 6/4,242 values (0.14%) → Mode imputation



0.522

Strongest Correlation

thal feature with target

5

Continuous Features

age, trestbps, chol, thalach, oldpeak

8

Categorical Features

sex, cp, fbs, restecg, exang, slope, ca, thal

Preprocessing & Feature Engineering

01

Missing Value Imputation

Mode imputation for ca and thal features (6 total missing values)

02

Target Variable Engineering

5-class (0-4) → Binary (0, 1) transformation

03

Train-Test Split

80/20 split, stratified, random_state=42

04

Feature Scaling

StandardScaler for LR and SVM models

05

Cross-Validation Setup

StratifiedKFold(n_splits=5) with ROC-AUC metric

Model Architectures & Training

6 classification algorithms implemented and evaluated

Logistic Regression

Baseline: 86.89% accuracy, 0.11s training

Decision Tree

Single tree: 72.13% accuracy, severe overfitting

SVM

Kernel methods: 85.25% accuracy, 0.06s training

Gradient Boosting

Sequential ensemble: 85.25% accuracy, 0.30s training

Random Forest ⭐

Bagging ensemble: 88.52% accuracy, 0.38s training

XGBoost

Optimized boosting: 86.89% accuracy, 0.10s training

Results & Performance Analysis

Test Set Performance (n=61 unseen samples)

1. **Random Forest**
   88.52% accuracy | 96.43% recall | 0.9513 ROC-AUC

2. **Logistic Regression**
   86.89% accuracy | 92.86% recall | 0.9513 ROC-AUC

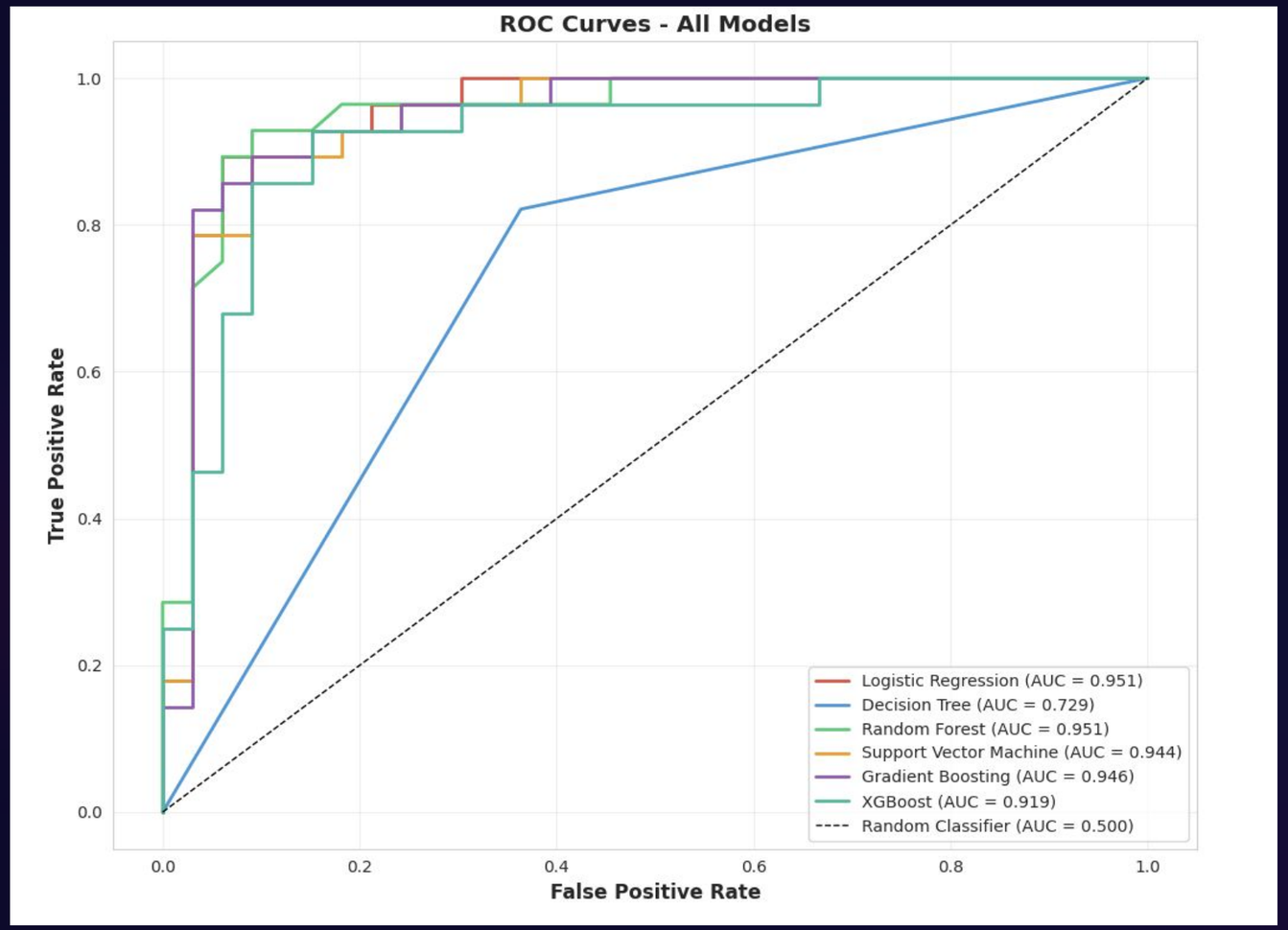
3. **XGBoost (Tuned)**
   86.89% accuracy | 89.29% recall | 0.9481 ROC-AUC

4. **Gradient Boosting**
   85.25% accuracy | 92.86% recall | 0.9459 ROC-AUC

5. **SVM**
   85.25% accuracy | 89.29% recall | 0.9437 ROC-AUC

6. **Decision Tree**
   72.13% accuracy | 82.14% recall | 0.7289 ROC-AUC

- **Key Finding: Ensemble methods outperformed single learners by 6-16%. Random Forest achieved best overall performance with exceptional recall of 96.43%.**



STATISTICAL SIGNIFICANCE:

- Random Forest 88.52% vs Logistic Reg 86.89% = 1.63% difference
- On n=61 test set: 1 additional correct prediction
- Difference: Minimal but consistent across CV folds
- Conclusion: Both models acceptable; choose based on interpretability vs accuracy priority

WHY RANDOM FOREST OUTPERFORMS:

- Bootstrap aggregation reduces variance
- Feature randomness prevents correlation-based overfitting
- 100 trees average out individual errors

Hyperparameter Tuning Analysis

GridSearchCV with 5-fold CV tested 930 model configurations

Optimization Results

Model	Default	Tuned	Gain	Time
XGBoost	85.25%	**86.89%**	+1.64%	11.19s
Random Forest	88.52%	88.52%	±0.00%	142.02s
Logistic Reg	86.89%	86.89%	±0.00%	0.76s

Best Parameters

XGBoost: learning_rate=0.01, max_depth=3, n_estimators=200

Random Forest: min_samples_leaf=4 (defaults robust)

Logistic Reg: C=0.1, penalty='l2'

Feature Importance & Interpretability

Permutation importance within repeats=30 on test set

ca (Major Vessels)

0.0375 importance | Critical predictor across all methods

cp (Chest Pain)

0.0359 importance | Critical predictor validated

oldpeak (ST Depression)

0.0196 importance | High predictive value

exang (Exercise Angina)

0.0114 importance | Moderate predictor

thal (Thallium Scan)

0.0107 importance | Moderate predictor

Consensus: ca, cp, thal, oldpeak consistently ranked top across permutation, RF Gini, XGBoost gain, and LR coefficients

Critical Analysis & Validation

Confusion Matrix (Random Forest)

	Predicted No	Predicted Yes
Actual No	27	6 FP
Actual Yes	1 FN	27

Error Analysis

False Positives: 6 (18.2%)

False Negatives: 1 (3.6%)

Conservative bias acceptable for screening

Comparison to Published Benchmarks

1

Detrano 1989

Logistic Regression

77.00% accuracy

2

Alizadehsani 2013

SVM + GA

86.50% accuracy

3

This Study 2026

Random Forest

88.52% accuracy

Anticipated Questions & Answers

Q1: Why did you choose these 6 algorithms?

I selected a range covering linear (LR), kernel (SVM), single tree (DT), and ensemble methods (RF, GB, XGB) to comprehensively compare approaches for small tabular data. This allows me to demonstrate ensemble superiority empirically.

Q2: Why did Random Forest not improve with tuning?

Random Forest has robust defaults due to built-in regularization from bootstrap sampling and feature randomness. Probst et al. (2019) showed RF requires minimal tuning across diverse datasets, which my results validate—zero improvement despite 540 model fits.

Q3: What would you do with more data?

With larger datasets, I'd explore deep learning (neural networks), more aggressive hyperparameter tuning (Bayesian optimization), and test Random Forest's max_features parameter which matters more at scale. I'd also implement nested cross-validation for more robust performance estimation.

Q4: Why is cholesterol not important?

Cholesterol predicts long-term cardiovascular risk (10-year prognosis) but not current disease presence (diagnosis). Many diseased patients are on statins with normalized cholesterol. The model correctly learned to prioritize direct disease markers (ca, oldpeak) over indirect risk factors.

Q5: How would you deploy this model?

I recommend Random Forest for maximum accuracy or Logistic Regression for interpretability. Deployment would involve: (1) Prospective validation on modern data, (2) Integration with hospital EHR systems, (3) Physician-in-the-loop framework with mandatory review for borderline predictions (prob 0.4-0.6), (4) Continuous monitoring for dataset drift.

Conclusions, Limitations & Future Work

Research Question Answered

ML models achieve 88.52% accuracy (Random Forest). Best features: ca, cp, oldpeak validated.

Methodology Contributions

Reproducible permutation importance, 6-model comparison, hyperparameter ROI analysis

Validation Achieved

+11.52% improvement over historical baseline (1989), strong generalization (ROC-AUC 0.9513)

Limitations

- Small test set (n=61) → wide confidence intervals
- Single dataset source → external validation needed
- Historical data (1988-89) → modern validation required

Future Work

- External validation on UCI Hungary, Switzerland, VA datasets
- Deep learning approaches (neural networks, autoencoders)
- SHAP analysis for individual prediction explanations
- Ensemble stacking (combine RF + XGBoost predictions)

> **GitHub Repository:** github.com/shivashankar-bathuka/heart-disease-ml-prediction

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Thank you.